Mohamed-Yassine Hemissi

AI Engineering Student — GenAI, LLM Applications & Applied Deep Learning

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Profile

Computer science has been my passion for as long as I can remember, driven by a curiosity for how things work and how technology can change lives. From building software for clients to co-founding a business and exploring AI research, I've always been drawn to turning ideas into something real. Now completing my AI engineering degree, I aspire to pursue research and entrepreneurship, contributing to advances in AI while building technology that makes a meaningful impact.

Education

ESPRIT — Engineering School (EUR-ACE) | *AI Engineering Degree — 16/20, honours track* *Tunis* | **2024 – 2027 (exp.)**

Deep learning, applied AI and data-driven systems, probability and Markov methods, linear optimization, graph theory.

University of Tunis El Manar — Faculty of Sciences | *B.Sc. Computer Science — 15/20, top 10%* *Tunis* | **2021 – 2024**

Professional Experience

ODDO BHF | *AI Engineering Intern* | **Jun 2026 – Aug 2026**

- Delivered three GenAI automations on **Prisme.ai** (enterprise agent platform) for the OBAM Asset Management Data team, each from problem statement to production.
- Automated document extraction and form filling, and Excel/PDF extraction into populated Excel and PDF outputs, at **95% accuracy**; cut runtime on the slowest process by **25×**.
- Built a Word-document automation (cleaning, layout, extraction, filling) at **80–90% user satisfaction**; evaluated all three in UAT with business users over **60 cases across 5 expert-reviewed batches**.

ESPRIT — Mobility Department | *AI Engineering Intern (part-time)* | **Mar 2026 – May 2026**

- In a team of 3, built LLM conversational assistants (**LangChain**, LangGraph) with **Langfuse** tracing, automated form processing and their internal platforms, cutting departmental repetitive workload by **50–60%**.

H&H Agency | *Co-Founder & Software Engineer (part-time)* | **2021 – 2026**

- Co-founded a software agency and delivered **15+** web and backend projects across France, Tunisia, Qatar, the USA and Canada, owning scoping, architecture and delivery.

Artify Inc | *Software Engineering Intern (two terms)* | **Jan – Jul 2024 & Jun – Aug 2025**

- Built the full backend from scratch for a producer content-management platform: ingestion pipelines, database design and the APIs feeding Artify's streaming service.
- Shipped content-management features with the CEO, enabling onboarding of **100+** media producers.

Sagemcom | *Software Development Intern* | **Jun 2023 – Aug 2023**

- Built internal tooling automating KPI discovery and production-data tracing, cutting BI dashboard prep time for factory engineers.

Bubble Fest | *Chief Technology Officer (CTO)* | **Apr 2022 – Jul 2022**

- Designed and deployed ticketing infrastructure handling **30,000+ requests** during peak ticket sales, while reducing technology costs by approximately **95%** through an open-source stack.

Selected AI Projects

DeglaAI — Early Palm Tree Disease Detection (Deep Learning + IoT) — **Top 5**, APII Incubator Competition **2025**

PyTorch, Vision Transformer, ResNet, spectrogram audio, Arduino

- Fine-tuned a **Vision Transformer** on spectrograms of palm-grove audio to detect pest-related acoustic anomalies, trained on ~**3,000 clips** skewed towards normal recordings, reaching **95% accuracy** on a held-out test set; paired with an Arduino capture unit (microphone, LED) streaming audio to the inference service.
- Fine-tuned a **ResNet CNN** for leaf-disease detection, fusing and cleaning several public datasets into one consistent labelled set; integrated both models into a palm-farming ERP.

Metabolic Health Decision-Support System (Insulin Resistance) **2025**

Python, scikit-learn, regression modelling, RAG, semantic scoring

- Trained a regression model estimating the **METS-IR** insulin-resistance index **without any lab values**, from inputs measurable at home, where the reference formula requires fasting glucose, triglycerides and HDL.
- Combined it with a **weighted semantic symptom-scoring** classifier for explainable, non-diagnostic output; built a nutritional pipeline into a structured Tunisian food dataset used as **RAG** context for dietary advice, documented in a short paper.

TMS — AI Platform for the Tunisian Poultry Industry — **2nd of 100+**, SDG Award, Bal des Projets ESPRIT **2026**

LLM prompting & agent design, Python, workflow automation

- Built the **financial assistant module** of a team platform for AI-driven poultry farm operations, giving users conversational access to their financial documents.
- Built an automation generating synthetic financial documents, used to validate the finance analytics.

Technical Skills

GenAI & Agents: LangGraph, LangChain, OpenAI SDK, RAG, LLM agents, LLM/RAG evaluation, Langfuse, Prisme.ai, Qdrant, Chroma
ML & Deep Learning: PyTorch, fine-tuning, Vision Transformers, ResNet, audio/spectrogram pipelines, scikit-learn, pandas, NumPy
Backend & Data: Python, TypeScript, FastAPI, NestJS, Node.js, SQL, MongoDB, Docker, AWS (Lambda, S3)
Languages: French (Native) | Arabic (Native) | English (C1)
Certifications: IBM AI Enterprise Workflow L2 | DataCamp Associate AI Engineer | AWS ML Foundations | NVIDIA DLI

Awards, Hackathons & Activities

1st place — H12 Hackathon (2026): *Gabbiest*, a CSR platform with LangGraph audit agents. | **2nd of 40** — Lunar Hackathon (2026): *Smartini*, LLM sales agents for SMEs.

Programming Instructor (2022–2025) — taught core computer science to cohorts of 15–20 students.